Polynomial-Time Approximation of the Star Height of a DFA

Yoav Danieli

1 Regular expressions and star height

Fix a finite alphabet Σ.

Definition 1 (Ordinary regular expression)**.** *The regular expressions over Σ are generated by*

$$E ::= \emptyset \mid \varepsilon \mid a \mid E \cup E \mid E \cdot E \mid E^*, \qquad a \in \Sigma.$$

Their semantics is

$$\begin{aligned} L(\emptyset) &= \emptyset, & L(\varepsilon) &= \{\varepsilon\}, & L(a) &= \{a\}, \\ L(E \cup F) &= L(E) \cup L(F), & L(EF) &= \{uv : u \in L(E),\, v \in L(F)\}, \\ L(E^*) &= \bigcup_{m \geq 0} L(E)^m. \end{aligned}$$

Definition 2 (Star height)**.** *The star height $h(E)$ of a regular expression is defined by*

$$\begin{aligned} h(\emptyset) = h(\varepsilon) = h(a) &= 0, \\ h(E \cup F) = h(EF) &= \max\{h(E), h(F)\}, \\ h(E^*) &= 1 + h(E). \end{aligned}$$

For a regular language $L \subseteq \Sigma^*$, its star height is

$$h(L) = \min\{h(E) : L(E) = L\}.$$

Since expressions without Kleene star denote finite languages,

$$h(L) = 0 \iff L \text{ is finite.}$$

For a DFA, finiteness is decidable in polynomial time by checking whether a directed cycle is reachable from the initial state and can reach an accepting state.

2 DFA input model

Definition 3 (Deterministic finite automaton)**.** *A DFA is a tuple*

$$A = (Q, \Sigma, \delta, q_0, F),$$

where Q is finite, $\delta : Q \times \Sigma \to Q$ is total, $q_0 \in Q$, and $F \subseteq Q$. Its language is

$$L(A) = \{w \in \Sigma^* : \delta(q_0, w) \in F\}.$$

The input size parameter used below is $n = |Q|$. Unless stated otherwise, the DFA is assumed to be trim and minimal.

3 Polynomial-size output: regular-expression circuits

An explicit regular expression equivalent to an n-state DFA may require exponential size. A polynomial-time algorithm therefore cannot always print the fully unfolded expression. We use shared subexpressions.

Definition 4 (Regular-expression circuit)**.** *A regular-expression circuit is a finite directed acyclic graph with one output node. Each node is labelled by*

$$\emptyset, \quad \varepsilon, \quad a \in \Sigma, \quad \cup, \quad \cdot, \quad \textit{or} \quad *,$$

with the expected arity. Its semantics is obtained by evaluating the circuit as a regular expression. Its size is its number of nodes.

The star height of the circuit is the maximum number of $*$-nodes on a directed path from the output to a leaf. This equals the star height of its unfolding.

A *constructive polynomial-time approximation algorithm* must output a polynomial-size regular-expression circuit. An output-sensitive variant may instead print an ordinary expression in time polynomial in the input and output sizes.

4 Kleene's dynamic-programming construction

Let $Q = \{q_1, \dots, q_n\}$. For $0 \le k \le n$, let $R_{ij}^{(k)}$ denote the words taking q_i to q_j whose intermediate states belong to $\{q_1, \dots, q_k\}$. The standard recurrence is

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} \cup R_{ik}^{(k-1)} \big(R_{kk}^{(k-1)}\big)^* R_{kj}^{(k-1)}.$$

The base expressions $R_{ij}^{(0)}$ are unions of transition labels, together with ε when $i = j$. The output is the union of $R_{0f}^{(n)}$ over accepting states f.

With sharing, this gives a polynomial-size circuit. Syntactically,

$$h\big(R_{ij}^{(k)}\big) \le k,$$

up to an inessential indexing shift. Thus the construction gives height at most n. For every infinite language, $h(L) \ge 1$, so this is an $O(n)$-factor constructive approximation.

The order of the states can affect the resulting expression substantially, but merely choosing the best ordering of the input DFA does not necessarily find the optimal star height of the language.

5 Cycle rank and the automata characterization

Definition 5 (Cycle rank)**.** *Let G be a finite directed graph. Its cycle rank $r(G)$ is defined recursively:*

(i) if G is acyclic, then $r(G) = 0$;

(ii) if G is strongly connected and has at least one edge, then

$$r(G) = 1 + \min_{v \in V(G)} r(G - v);$$

(iii) otherwise, $r(G)$ is the maximum cycle rank of a strongly connected component of G.

For an automaton B, write $r(B)$ for the cycle rank of its transition graph.

Theorem 1 (Eggan)**.** *For every regular language L,*

$$h(L) = \min\{r(B) : B \text{ is an NFA accepting } L\}.$$

In particular, for the input DFA A,

$$h(L(A)) \le r(A) \le n.$$

The first inequality may be strict because Eggan's minimum ranges over all equivalent NFAs, not merely the input DFA or its state-elimination order.

Definition 6 (Bideterministic automaton)**.** *A trim DFA is bideterministic if it has one accepting state and reversing all transitions, while exchanging initial and accepting states, yields a deterministic automaton.*

For bideterministic languages, the minimal trim DFA satisfies

$$h(L(A)) = r(A).$$

Consequently, approximation of star height for this class is exactly approximation of cycle rank.

6 Approximation conventions

A multiplicative approximation needs special treatment at optimum 0.

Definition 7 (Value approximation)**.** *Let $\alpha : \mathbb{N} \to [1, \infty)$. A polynomial-time $\alpha(n)$-approximation for DFA star height is an algorithm which, on input an n-state minimal DFA A, outputs an integer $\widehat{h}(A)$ such that:*

(i) if $h(L(A)) = 0$, then $\widehat{h}(A) = 0$;

(ii) if $h(L(A)) \ge 1$, then
$$h(L(A)) \le \widehat{h}(A) \le \alpha(n)\, h(L(A)).$$

Definition 8 (Constructive approximation)**.** *A constructive $\alpha(n)$-approximation outputs a polynomial-size regular-expression circuit C_A satisfying*

$$L(C_A) = L(A),$$

and
$$\begin{cases} h(C_A) = 0, & h(L(A)) = 0, \\ h(C_A) \le \alpha(n)\, h(L(A)), & h(L(A)) \ge 1. \end{cases}$$

Constructive approximation is stronger than value approximation because the output circuit certifies the upper bound.

Definition 9 (Best polynomial-time approximation ratio)**.** *Let $\alpha^*_{\mathrm{val}}(n)$ be the pointwise infimum, up to asymptotic constants, of the approximation ratios achieved by polynomial-time value algorithms. Define $\alpha^*_{\mathrm{con}}(n)$ analogously for constructive algorithms.*

The central quantitative problem is to determine the asymptotic growth of these functions.

The Kleene construction gives

$$\alpha^*_{\mathrm{con}}(n) \le n.$$

For bideterministic input, known cycle-rank algorithms give a polynomial-time

$$O\big((\log n)^{3/2}\big)$$

approximation.

7 Moderate conjecture

Conjecture 1 (Moderate I: constant approximation for bideterministic DFAs)**.** *There exists a universal constant c and a polynomial-time algorithm which, given a minimal bideterministic DFA A, outputs an equivalent regular-expression circuit C satisfying*

$$h(C) \leq c\, h(L(A)).$$

Because $h(L(A)) = r(A)$ in this class, this is equivalent to a constant-factor approximation for cycle rank on the transition graphs of bideterministic automata. It would improve the known $O((\log n)^{3/2})$ ratio.

References

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